Galaba Vamsi

Robotics Software Engineer, Intern / New Grad (graduating May 2027)
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SUMMARY

Mechatronics at IIT Bhilai. I build tooling that keeps robot data and robot changes honest: 15 merged PRs (11 code) to open-source robotics tools, 10 of them at two YC S26 companies; I maintain a PyPI plugin and co-founded Robotropic. Simulation and controls in ROS 2 (ROS2), Gazebo, MuJoCo, and MATLAB/Simulink.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology (IIT) Bhilai Aug 2023 - May 2027
Coursework: Optimal Control, Advanced Control Theory, Deep Learning for Computer Vision, AI and ML, Digital Image Processing, DSA, OOP.

OPEN-SOURCE CONTRIBUTIONS

HFlow, Hebbian Robotics (YC S26) github.com/Hebbian-Robotics/hflow Sep 2026 - Oct 2026
- 8 merged PRs (Python, Apache Arrow) to an SDK that verifies robot training data: LeRobot and robot-log conversion now keeps data whole or refuses it with a reason. Blocked silent 12- to 10-bit depth loss (PR #400), typed Arrow columns from every message (PR #657), fixed video-cache collisions (PR #353).

Inspect Robots, Robocurve (YC S26) github.com/robocurve/inspect-robots Sep 2026 - Oct 2026
- Evaluation framework for embodied-AI policies. 2 merged PRs: a configurable LLM-API retry policy across 5 wire protocols (PR #476) and unknown git state no longer logged as clean (PR #475); 3 open, including checkpoint/resume for long eval sets (PR #514).
- Wrote and maintain inspect-robots-wandb, a Weights & Biases plugin (pypi.org/project/inspect-robots-wandb): 100% coverage gate, CI on Python 3.10-3.13, trusted publishing.

Cerulion, Rust robot runtime 1 merged CLI fix (PR #216), 4 documentation fixes | github.com/cerulion-inc/cerulion Oct 2026

EXPERIENCE

Co-founder, Robotropic Physical-AI startup, with Varun Rao | robotropic.com Oct 2026 - Present
- Building Harness, a workspace where a factory's team and AI agents change robot skills safely: each change is checked against safety limits, rolled out cell by cell, and recorded. I lead the simulated factory floor and the change gateway.

Research Intern, IIT Bhilai Advisor: Dr. Arzad Alam Kherani | COMET Foundation project May 2025 - Jul 2025
- Built RIS-SIM, an open-source Python emulator for reconfigurable intelligent surface (RIS) radio links with USRP-style SDR interfaces; designed RIS unit cells in CST Studio Suite (IEEE INDICON 2025, co-author). Rebuilt it in 2026 as RIS-SIM v2 (103 tests), then as sremu.

Co-founder, Human Slop Anti-AI social app, with Varun Rao | humanslop.in Mar 2026 - Present
- Expo (React Native) app for web and Android, Fastify API, PostgreSQL on Supabase; live on the web and Google Play (50+ downloads).

PROJECTS

Quarter-Car Active Suspension (course project, MATLAB/Simulink, 2026) Reproduced Na et al. (IEEE TSMC 2022) to within about 6% of its error indices after fitting 2 plant parameters, then showed its gains fail a hold-out test, its zero-order-hold versions go unstable through an 88 Hz mode, and LQR and skyhook match it at equal effort. Kalman filter, anti-windup extension, C port timed for a Cortex-M4 (QEMU). github.com/Galabavamsi/quarter-car-afc-replication

Whiteboard Cleaning Robot (team of 2, ROS 2 Jazzy, Gazebo Harmonic, 2025) 3-DOF Cartesian gantry: URDF/Xacro, ros2_control, rclpy, a raster coverage path (10 cm overlap), logged coverage analysis. github.com/Galabavamsi/whiteboard_robot

Pehchaan (team of 2; Top 10, RAKSHAM (IIT Delhi, I4C and Amazon), 2026) Voice-clone payment-fraud defence; I led ML and backend. Sim-to-real: a detector at 4.8% equal error rate (EER) on simulated channels hit 24.6% on 48 live carrier calls; modelling the real speakerphone channel and retraining cut it to 6.8% with 0% of genuine callers flagged. github.com/Galabavamsi/pehchaan

sremu (2026, MIT, built with Claude Code) Sample-accurate IQ-level RIS emulator for controller-in-the-loop tests: timestamped commands act on an exact, logged sample; deterministic replay; runs unmodified GNU Radio 3.10 over ZeroMQ; checked against analytic references and MATLAB, bit-identical on Windows and Linux. github.com/Galabavamsi/sremu

SCARA Pick-and-Place (MuJoCo, 2026) SCARA arm with analytic inverse kinematics; a text command such as "sort red, blue, green" is parsed (rule-based) into a pick-and-place plan. github.com/Galabavamsi/prompt-robot-arm

SKILLS

Robotics and controls: ROS 2 (rclpy, ros2_control, URDF/Xacro), Gazebo Harmonic, MuJoCo, MATLAB/Simulink, LQR, Kalman filter, inverse kinematics, LeRobot datasets, Unity
Languages: Python, C (embedded, Cortex-M4), C++, Rust, TypeScript, SQL
Tools: Git, Linux, Docker, GitHub Actions, pytest, PyTorch, Apache Arrow, Weights & Biases, GNU Radio
AI-assisted development: Claude Code; I set the design and verify results with tests and independent references.

PUBLICATIONS AND PREPRINTS

- **Co-author**, *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306; accepted to the SPAIS Workshop at CoRL 2026 (poster).
- **Co-author**, 2 RIS papers: IEEE INDICON 2025 (RF energy-harvesting RIS) and *Ad Hoc Networks* (Elsevier), 2026 (systematic review of RIS simulation models).
- **Sole author**, *Timed RIS Control in the Loop*, preprint draft, arXiv submission pending (github.com/Galabavamsi/sremu-paper).

ACHIEVEMENTS

- **Top 10 (Grand Jury round) of 4,500+ teams nationwide**, RAKSHAM AI Cybersecurity Hackathon (IIT Delhi, I4C and Amazon), Oct 2026; Pehchaan, team of 2.
- **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team, Dec 2025); **5th of 232 teams**, WEAR Dataset Challenge @HASCA 2026 (Kaggle); **All India Rank 200**, Amazon ML Challenge 2026 (team).
- Prime Minister's Scholarship Scheme (PMSS) scholar, 2024; Subroto Mukherjee Scholarship (IAFBA), 2023; Top 1% in JEE Advanced; attended YC Startup School, Bengaluru.